

EXP NO: 3 Date: 6-8-26	Classification with Decision Trees
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AIM:

To implement a Decision Tree Classifier using the Iris dataset and evaluate its performance using Accuracy Score and Confusion Matrix.

ALGORITHM:

- Import the necessary Python libraries for data loading, model building, evaluation, and visualization.
- Load the Iris dataset from the `sklearn.datasets` module.
- Separate the dataset into input features (**X**) and target labels (**y**).
- Split the dataset into training and testing sets using `train_test_split()`.
- Create a Decision Tree Classifier using the **Gini Index** as the splitting criterion.
- Set a random state to ensure reproducible results.
- Train the Decision Tree model using the training dataset.
- Use the trained model to predict the class labels for the test dataset.
- Generate the Confusion Matrix to compare the predicted labels with the actual labels.
- Calculate the Accuracy Score to measure the overall performance of the classifier.
- Display the Confusion Matrix using a heatmap for better visualization.
- Visualize the trained Decision Tree to understand the decision-making process.
- Analyze the results and verify the classification performance of the model.
- Conclude the experiment based on the obtained accuracy and visualization results.

CODE:

```
from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.model_selection import train_test_split
from sklearn.metrics import confusion_matrix, accuracy_score
import matplotlib.pyplot as plt
import seaborn as sns

iris=load_iris()
X=iris.data
y=iris.target

X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.3,random_state=42)

dt_model=DecisionTreeClassifier(criterion='gini',random_state=42)
dt_model.fit(X_train,y_train)

y_pred=dt_model.predict(X_test)

cm=confusion_matrix(y_test,y_pred)
acc=accuracy_score(y_test,y_pred)
print("Confusion Matrix:\n",cm)
print("Accuracy Score:",acc)

sns.heatmap(cm,annot=True,cmap="Blues",xticklabels=iris.target_names,yticklabels=iris.target_names)
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()

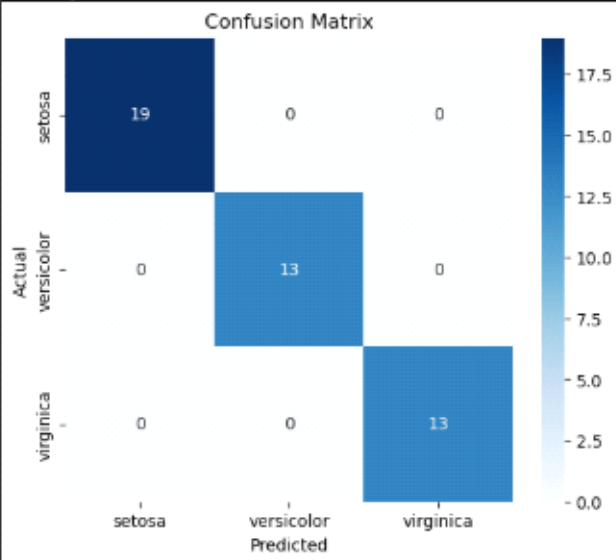
plt.figure(figsize=(12,8))
plot_tree(dt_model,filled=True,feature_names=iris.feature_names,class_names=iris.target_names)
plt.title("Decision Tree Visualization")
plt.show()
```

OUTPUT:

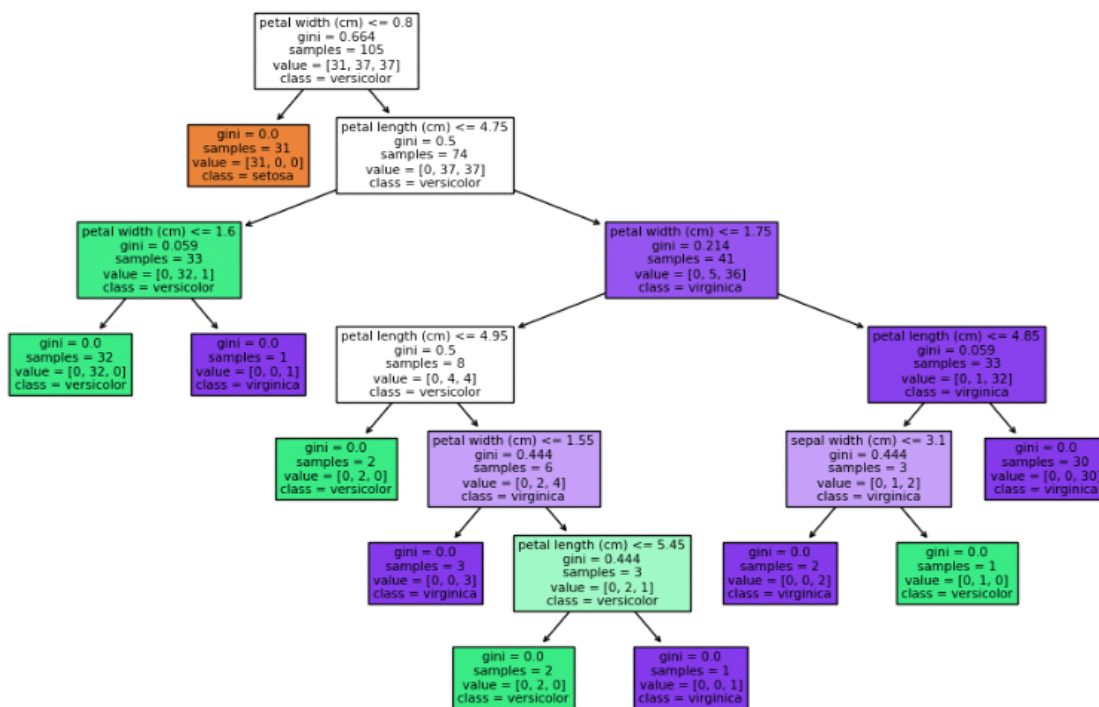
Confusion Matrix:

```
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
```

Accuracy Score: 1.0



Decision Tree Visualization



RESULT:

The Python environment was successfully set up and the dataset was pre-processed by handling missing values, encoding categorical data, performing feature scaling, and splitting the data into training and testing sets. The dataset is now ready for model training and analysis.